

CSP-Bender Problem

Formulation, Solution in AMPL, and Comparison with Heuristic Methods

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1 Introduction

Objectives:

- TODO

2 Problem Description and Formulation

2.1 Basic definitions

- **Microdata:** a file V of s individuals and t variables, as a matrix $s \times t$ where V_{ij} is the value of variable j for individual i .
- **Tabular data:** tables generated crossing one or more categorical variables from microdata. These can be multidimensional tables holding frequency or magnitude information. The table will have n cells which will have to verify m linear constraints. The cells in the table are bounded by an upper and lower limit: $u < a_i < l$
- **Protection levels from the p% rule:**

An attacker might estimate the intervals of a cell $[\bar{a}_p, \underline{a}_p]$, to assert these are not informative enough, we must define some lower and upper protection level bounds to ensure this interval fully contains the interval

$$[\bar{a}_p, \underline{a}_p] \in [a_p - lpl_p, a_p + upl_p] \quad (1)$$

2.2 CSP Formulation

The Cell Suppression Problem seeks to hide sensitive (primary, $\mathcal{P}$) cells while minimizing the suppression cost. Hiding only primary cells would not be enough since we could still infer information about them from the rest of cells, thus, a number of secondary cells ($\mathcal{S}$) will also be suppressed.

CSP computes the set $\mathcal{S}$ of secondary cells to be suppressed such that for all $p \in \mathcal{P}$, Equation 1 is satisfied, where $\bar{a}_p, \underline{a}_p$ can be obtained with a minimization problem.

The CSP problem can be formulated as a Mixed-Integer Linear Programming (MILP) problem. The MILP will be a minimization problem in terms of the following variables

- $y_i \in \{0, 1\}, i = 1, \dots, n$: Indicates if a cell i is suppressed (1) or not (0)
- $x^{l,p} \in \mathbb{R}^n$: Lower deviation of the primary cell p from its original value
- $x^{u,p} \in \mathbb{R}^n$: Upper deviation of the primary cell p from its original value

MILP formulation for CSP:

$$\begin{aligned} \min \quad & \sum_{i=1}^n w_i y_i \\ \text{s.to} \quad & \left. \begin{aligned} & Ax^{l,p} = 0 \\ (l_i - a_i)y_i \leq & \begin{cases} x_i^{l,p} \leq (u_i - a_i)y_i \\ x_p^{l,p} \leq -lpl_p \end{cases} & i = 1, \dots, n \end{aligned} \right\} \forall p \in \mathcal{P} \\ & \left. \begin{aligned} & Ax^{u,p} = 0 \\ (l_i - a_i)y_i \leq & \begin{cases} x_i^{u,p} \leq (u_i - a_i)y_i \\ x_p^{u,p} \geq upl_p \end{cases} & i = 1, \dots, n \end{aligned} \right\} \\ & y_i \in \{0, 1\} \quad x \geq 0 \quad i = 1, \dots, n, \end{aligned}$$

The problem with the MILP formulation is that it becomes infeasibly complex as the table grows large.

The problem has:

- n binary variables and $2n\mathcal{P}$ continuous variables
- $2(m + 2n)|\mathcal{P}|$ constraints

2.3 Benders decomposition

Benders decomposition partitions the problem into a master and subproblems, and iterates efficiently between them to find the optimal solution.

- Master problem: focuses on the binary decisions (y_i), which determine whether a cell is published ($y_i = 0$) or suppressed ($y_i = 1$)
- Subproblems: verify if the current suppression pattern chosen by the Master Problem actually provides the required protection levels (lpl and upl) for each primary sensitive cell.

Original Problem

Given the original problem:

$$\begin{aligned} (P) = \quad & \min \quad c^T x + q^T y \\ & \text{s. to} \quad Tx + Wy = h \\ & \quad \quad x \in X \\ & \quad \quad y \geq 0 \end{aligned}$$

The two following problems are derivated:

• Master Formulation

$$\begin{aligned} (BP_r) = \quad & \min \quad z \\ & \text{s. to} \quad z \geq c^T x + u^{i\top}(h - Tx) \quad i \in I \subseteq \{1 \dots p\} \\ & \quad \quad v^{j\top}(h - Tx) \leq 0 \quad j \in J \subseteq \{1 \dots q\} \\ & \quad \quad x \in X. \end{aligned}$$

• Subproblem Formulation

$$\begin{aligned} (Q_D) = \quad & \max \quad u^T(h - Tx) \\ & \text{s. to} \quad W^T u \leq q \\ & \quad \quad u \in \mathbb{R}^m \end{aligned}$$

To take advantage of Benders decomposition we first reformulate the original MILP problem to an equivalent of Benders starting problem.

$$\begin{aligned}
\min \quad & \sum_{i=1}^n c_i y_i \\
\text{s. to} \quad & Ax^{l,p} = 0 & \forall p \in \mathcal{P} \\
& Ax^{u,p} = 0 & \forall p \in \mathcal{P} \\
& (l_i - a_i)y_i - x_i^{l,p} + s_{l,i}^{l,p} = 0 & i = 1 \dots n, \forall p \in \mathcal{P} \\
& (a_i - u_i)y_i + x_i^{l,p} + s_{u,i}^{l,p} = 0 & i = 1 \dots n, \forall p \in \mathcal{P} \\
& (l_i - a_i)y_i - x_i^{u,p} + s_{l,i}^{u,p} = 0 & i = 1 \dots n, \forall p \in \mathcal{P} \\
& (a_i - u_i)y_i + x_i^{u,p} + s_{u,i}^{u,p} = 0 & i = 1 \dots n, \forall p \in \mathcal{P} \\
& x_p^{l,p} + s_p^{l,p} = -lpl_p & \forall p \in \mathcal{P} \\
& x_p^{u,p} - s_p^{u,p} = upl_p & \forall p \in \mathcal{P} \\
& y_i \in \{0, 1\}, \quad x, s \geq 0
\end{aligned}$$

Given that y_i belongs to a restricted set and both x and s are only restricted to be positive, we take y_i as the x in Benders and $[x, s]$ as the y .

The Original Problem now becomes:

Objective function (min)		$\mathbf{c}^T \mathbf{x}$	$\mathbf{q}^T \mathbf{y}$
		cy	$+0x^{l,p} + x^{u,p}$
Constraint Type	$\mathbf{Tx}$	$+\mathbf{Wy}$	$=\mathbf{h}$
Table Conservation	0	$+Ax^{l,p}$	$=b$
		$+Ax^{u,p}$	$=b$
Lower Capacity	$(l - a)y$	$-x^{l,p}$	≤ 0
	$(l - a)y$	$-x^{u,p}$	
Upper Capacity	$(a - u)y$	$+x^{l,p}$	≤ 0
	$(a - u)y$	$+x^{u,p}$	
Protection Levels	0	$x^{l,p}$	$\leq -lpl$
		$x^{u,p}$	$\geq upl$

$$\begin{aligned}
& x \in X & y \geq 0 \\
\text{Domain Constraints} & y_i \in \{0, 1\} & x, s \geq 0
\end{aligned}$$

Primal Subproblem Q: subproblem as an optimization to see if the protection level can be reached given the fixed suppression $\bar{y}$: for each primary cell $p[j]$:

$$\begin{aligned}
(Q_{lower}) = \quad & \min \quad x_{p[j]} \\
\text{s. to} \quad & Ax_l = 0[\lambda] \\
& (l_i - a_i)\bar{y}_i \leq x_{l,i}[\mu_{ll}] \\
& x_{l,i} \leq (u_i - a_i)\bar{y}_i[\mu_{ul}] \\
& \text{Goal : Objective value} \leq -lpl_{p[j]}?
\end{aligned}$$

$$\begin{aligned}
(Q_{upper}) = \quad & \max \quad x_{p[j]} \\
\text{s. to} \quad & Ax_u = 0[\lambda] \\
& (l_i - a_i)\bar{y}_i \leq x_{u,i}[\mu_{lu}] \\
& x_{u,i} \leq (u_i - a_i)\bar{y}_i[\mu_{uu}] \\
& \text{Goal : Objective value} \leq -upl_{p[j]}?
\end{aligned}$$

Dual of Subproblem Q: To ensure a suppression pattern is valid, you have to check if a primary cell is protected from "below" (someone trying to guess a value too low) and from "above" (someone trying to guess a value too high). Dual of Subproblem Q for each primary cell $p[j]$:

$$\begin{aligned}
(Q_{Dlower}) = \quad & \max \quad \sum_{i=1}^n ((l_i - a_i)\mu_{ll,i} - (u_i - a_i)\mu_{ul,i})\bar{y}_i \\
\text{s. to} \quad & A^T\lambda + \mu_{ll} - \mu_{ul} = e_p \\
& \mu_{ll} \geq 0, \mu_{ul} \geq 0, \lambda \text{ free}
\end{aligned}$$

If the optimal value of this objective is greater than $-lpl$, that means the pattern is insecure and we have to add a lower Benders cut to the Master Problem.

$$-lpl \geq \sum_{i=1}^n ((l_i - a_i)\mu_{ll,i} - (u_i - a_i)\mu_{ul,i})\bar{y}_i$$

$$\begin{aligned}
(Q_{Dupper}) = \quad & \max \quad \sum_{i=1}^n ((u_i - a_i)\mu_{uu,i} - (l_i - a_i)\mu_{lu,i})\bar{y}_i \\
\text{s. to} \quad & A^T\lambda + \mu_{lu} - \mu_{uu} = -e_p \\
& \mu_{lu} \geq 0, \mu_{uu} \geq 0, \lambda \text{ free}
\end{aligned}$$

If the optimal value of this objective is less than upl , that means the pattern is insecure and we have to add an upper Benders cut to the Master Problem.

$$upl \leq \sum_{i=1}^n ((u_i - a_i)\mu_{uu,i} - (l_i - a_i)\mu_{lu,i})\bar{y}_i$$

If we have P primary cells, we will have to solve $2P$ subproblems at each iteration of Benders decomposition.

Master Problem: We have to add the cuts for all the insecure primary cells found in the subproblems. If we have

- $\mathcal{J}_l$: set of lower cuts added until iteration r from insecure primary cells found with the lower dual subproblem
- $\mathcal{J}_u$: set of upper cuts added until iteration r from insecure primary cells found with the upper dual subproblem
- for each cut $j_l \in \mathcal{J}_l$, we have $\bar{\mu}_{ll}^{p,j_l}, \bar{\mu}_{ul}^{p,j_l}$ from the solution of the corresponding lower dual subproblem

- for each cut $j_u \in \mathcal{J}_u$, we have $\bar{\mu}_{lu}^{p,j_u}, \bar{\mu}_{uu}^{p,j_u}$ from the solution of the corresponding upper dual subproblem

The Master Problem at iteration r is:

$$\begin{aligned}
(BP_r) = \quad & \min \sum_{i=1}^n c_i y_i \\
\text{s. to} \quad & y_p = 1 \quad \forall p \in \mathcal{P} \\
& -lpl_p \geq \sum_{i=1}^n ((l_i - a_i) \bar{\mu}_{ll,i}^{p,j_l} - (u_i - a_i) \bar{\mu}_{ul,i}^{p,j_l}) y_i \quad \forall p \in \mathcal{P}, \forall j_l \in \mathcal{J}_l \\
& upl_p \leq \sum_{i=1}^n ((u_i - a_i) \bar{\mu}_{uu,i}^{p,j_u} - (l_i - a_i) \bar{\mu}_{lu,i}^{p,j_u}) y_i \quad \forall p \in \mathcal{P}, \forall j_u \in \mathcal{J}_u \\
& y_i \in \{0, 1\}
\end{aligned}$$

3 Datasets

TODO

4 Implementation Details

5 Computational Experiments and Results

5.1 small

Original table: In Table 1 we can see the original data table for *small.ampl* data instance created by rows($\rightarrow$), with the sensitive cells marked with *. Column 7 and Row 5 contain the totals for the rows and columns respectively.

	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7
Row 1	300*	8	5	5	5*	38	361
Row 2	8	68*	40	76	29	31	252
Row 3	11	33	20	60	35	44	203
Row 4	7	28	36*	41	22	63	197
Row 5	326	137	101	182	91	176	1013

Table 1: Original Small Instance Data Table with Sensitive Cells Marked with *

Number Cell	Lower Protection Level	Upper Protection Level
1	15	20
5	2	4
9	10	6
24	9	3

Table 2: Table of Small Instance Protection Levels for Sensitive Cells

Table Constraints:

Row Constraints:

1. $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 - x_7 = 0$

2. $x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} - x_{14} = 0$
3. $x_{15} + x_{16} + x_{17} + x_{18} + x_{19} + x_{20} - x_{21} = 0$
4. $x_{22} + x_{23} + x_{24} + x_{25} + x_{26} + x_{27} - x_{28} = 0$

Column Constraints:

5. $x_1 + x_8 + x_{15} + x_{22} - x_{29} = 0$
6. $x_2 + x_9 + x_{16} + x_{23} - x_{30} = 0$
7. $x_3 + x_{10} + x_{17} + x_{24} - x_{31} = 0$
8. $x_4 + x_{11} + x_{18} + x_{25} - x_{32} = 0$
9. $x_5 + x_{12} + x_{19} + x_{26} - x_{33} = 0$
10. $x_6 + x_{13} + x_{20} + x_{27} - x_{34} = 0$

5.1.1 Results

Cell ID	Cost	Is Primary
1	1	1
3	1	0
5	1	1
8	1	0
9	1	1
10	1	0
12	1	0
15	1	0
22	1	0
24	1	1

Table 3: Table of Cells to be Suppressed in Small Instance

	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7
Row 1	x	8	x	5	x	38	361
Row 2	x	x	x	76	x	31	252
Row 3	x	33	20	60	35	44	203
Row 4	x	28	x	41	22	63	197
Row 5	326	137	101	182	91	176	1013

Table 4: Small Data Table with Suppressing Sensitive Cells

5.2 example_2D

In Table 5 we can see the original data table for *example_2D.ampl* data instance created by rows($\rightarrow$), with the sensitive cells marked with *. Column 1 and Row 1 represent the totals for their respective dimensions.

Table Constraints:

Grand Total Consistency:

	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
Row 1	3220	84	632	930	843	731
Row 2	1529	3	336	309	484	397
Row 3	484	25	3	393*	48	15
Row 4	392	1	2	137*	145	107
Row 5	815	55	291*	91	166	212*

Table 5: Original Example_2D Instance Data Table with Sensitive Cells Marked with *

Number Cell	Lower Protection Level	Upper Protection Level
16	40	30
22	14	14
27	15	30
30	21	21

Table 6: Table of Example_2D Instance Protection Levels for Sensitive Cells

1. $x_7 + x_{13} + x_{19} + x_{25} - x_1 = 0$
2. $x_2 + x_3 + x_4 + x_5 + x_6 - x_1 = 0$

Column Constraints:

3. $x_8 + x_{14} + x_{20} + x_{26} - x_2 = 0$ *(Column 1)*
4. $x_9 + x_{15} + x_{21} + x_{27} - x_3 = 0$ *(Column 2)*
5. $x_{10} + x_{16} + x_{22} + x_{28} - x_4 = 0$ *(Column 3)*
6. $x_{11} + x_{17} + x_{23} + x_{29} - x_5 = 0$ *(Column 4)*
7. $x_{12} + x_{18} + x_{24} + x_{30} - x_6 = 0$ *(Column 5)*

Row Constraints:

8. $x_8 + x_9 + x_{10} + x_{11} + x_{12} - x_7 = 0$ *(Row 1)*
9. $x_{14} + x_{15} + x_{16} + x_{17} + x_{18} - x_{13} = 0$ *(Row 2)*
10. $x_{20} + x_{21} + x_{22} + x_{23} + x_{24} - x_{19} = 0$ *(Row 3)*
11. $x_{26} + x_{27} + x_{28} + x_{29} + x_{30} - x_{25} = 0$ *(Row 4)*

5.2.1 Results

5.3 targus

In Table 9 we can see the original data table for *targus.ampl* data instance created by rows($\rightarrow$), with the sensitive cells marked with *. Due to the large values in the table, we have used commas to separate thousands. Due to the row-wise size of the table, we have transposed the original table for better visualization. Row 1 contains the totals for the columns.

Table Constraints:

Row Constraints:

1. $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 - x_7 = 0$

Column Constraints:

2. $x_1 + x_8 + x_{15} + x_{22} - x_{29} = 0$

Cell ID	Cost	Is Primary
3	0	0
6	0	0
13	0	0
16	0	1
19	0	0
22	0	1
27	0	1
30	0	1

Table 7: Table of Cells to be Suppressed in example_2D Instance

	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
Row 1	3220	84	x	930	843	x
Row 2	1529	3	336	309	484	397
Row 3	x	25	3	x	48	15
Row 4	x	1	2	x	145	107
Row 5	815	55	x	91	166	x

Table 8: example_2D Data Table with Suppressing Sensitive Cells

5.3.1 Results

6 References

	Row 1	Row 2	Row 3	Row 4	Row 5	Row 6	Row 7	Row 8	Row
Col 1	16,847,261.8	20*	25	2,711,808	2,320,534	2,505,042.6	2,799,074.3	6,510,758	0
Col 2	4,373,279	5*	5*	719,049	659,680	688,962	756,529	1,549,049	0
Col 3	1,896,129	5*	5*	398,082	348,039	354,711	418,778	486,529	0
Col 4	1,808,861	0	5*	223,990	221,332	241,913	258,233	863,393	0
Col 5	578,289	0	96,897	30,909	92,338	79,518	219,127	0	0
Col 6	3,703,898	15*	5*	642,238	515,003	534,147	620,392	1,392,096	0
Col 7	124,336	5*	0	36,311	32,132	25,770	18,150	11,868*	0
Col 8	526,279	0	0*	93,589	94,957	110,930	81,799	145,004	0
Col 9	2,234,995	10*	5*	345,803	251,858	251,188	303,377	1,086,254	0
Col 10	818,286	0	0	166,535	136,556	146,259	217,066	151,870	0
Col 11	4,576,115.8	0	0	648,972	543,570	663,896.6	775,132.3	1,944,545	0
Col 12	485,328	0	0	63,767	75,442	87,305	59,953	198,859	0
Col 13	3,664,559.8	0	0	537,911	430,851	515,019.8	643,762.3	1,537,016	0
Col 14	426,230	0	0	47,294	37,277	51,672	71,417	208,870	0
Col 15	4,183,971	0	15*	701,549	602,281	618,037	647,021	1,625,068	0
Col 16	2,752,743	0	15*	488,613	392,395	363,490	402,925	1,105,305	0
Col 17	1,441,228	0	0	212,936	209,886	254,547	244,096	519,783	0
Col 18	0	0	0	0	0	0	0	0	0

Table 9: Original Targus Instance Data Table with Sensitive Cells Marked with *

A AMPL Files

```

1 param n; # observations
2 param k; # clusters
3 set M:={1..n};
4
5 param d{M,M};
6 var x{M,M} binary;
7
8
9 minimize f: sum{i in M}
10    sum{j in M} d[i,j]*x[i,j];
11
12 subject to one_cluster{i in M}:
13    sum{j in M} x[i,j] = 1;
14
15 subject to k_clusters:
16    sum{j in M} x[j,j] = k;
17
18 subject to cluster_belongs{i in M, j in M}:
19    x[j,j] >= x[i,j];

```

Listing 1: kmedoid.mod

```

1 # Reset enviroment
2 reset;
3
4 # Load model and data
5 model kmedoids.mod;
6 data kmedoids.dat;
7
8 option solver cplex;
9 solve;
10
11 display _solve_elapsed_time;

```

Number Cell	Lower Protection Level	Upper Protection Level
19	6	6
20	1.67	1.67
21	1.67	1.67
24	5	5
25	1.67	1.67
27	3.33	3.33
38	1.67	1.67
39	1.67	1.67
42	1.67	1.67
45	1.67	1.67
51	5	5
52	5	5
133	3480	3480

Table 10: Table of Targus Instance Protection Levels for Sensitive Cells

```

12
13 printf {i in M, j in M} "%s,%s,%i\n", i, j, x[i,j] > "x_matrix.csv";

```

Listing 2: kmedoid.run

```

1 def generate_dat(N, D, k, filename='AMPL/kmedoid.dat'):
2     with open(filename, 'w') as f:
3         f.write("#Define sets\n")
4         f.write(f"param n := {N};\n")
5         f.write(f"param k := {k};\n")
6
7         f.write("#Define parameters\n")
8         f.write(f"param d: {'\t'.join([str(j+1) for j in range(N)])} :=\n")
9         for i in range(N):
10            f.write(f" {i+1} " + ' '.join(f"{D[i][j]:.6f}" for j in range(N)) +
11            "\n")
12            f.write(";\n")
13

```

Listing 3: Python function to generate kmedoid.dat

Cell ID	Cost	Is Primary
19	1	1
20	1	1
21	1	1
24	1	1
25	1	1
27	1	1
37	1	0
38	1	1
39	1	1
42	1	1
45	1	1
51	1	1
52	1	1
78	611	0
87	715	0
99	298	0
106	432	0
133	14	1
134	172	0

Table 11: Table of Cells to be Suppressed in Targus Instance

	Row 1	Row 2	Row 3	Row 4	Row 5	Row 6	Row 7	Row 8	Row
Col 1	16,847,261.8	x	25	2,711,808	2,320,534	2,505,042.6	2,799,074.3	6,510,758	0
Col 2	4,373,279	x	5*	719,049	659,680	688,962	756,529	1,549,049	0
Col 3	1,896,129	x	5*	398,082	348,039	354,711	418,778	486,529	0
Col 4	1,808,861	0*	5*	223,990	221,332	241,913	258,233	863,393	0
Col 5	578,289	0	96,897	30,909	92,338	79,518	219,127	0	0
Col 6	3,703,898	15*	5*	642,238	515,003	534,147	620,392	1,392,096	0
Col 7	124,336	5*	0	36,311	32,132	25,770	18,150	11,868*	0
Col 8	526,279	0	0*	93,589	94,957	110,930	81,799	145,004	0
Col 9	2,234,995	10*	5*	345,803	251,858	251,188	303,377	1,086,254	0
Col 10	818,286	0	0	166,535	136,556	146,259	217,066	151,870	0
Col 11	4,576,115.8	0	0	648,972	543,570	663,896.6	775,132.3	1,944,545	0
Col 12	485,328	0	0	63,767	75,442	87,305	59,953	198,859	0
Col 13	3,664,559.8	0	0	537,911	430,851	515,019.8	643,762.3	1,537,016	0
Col 14	426,230	0	0	47,294	37,277	51,672	71,417	208,870	0
Col 15	4,183,971	0	15*	701,549	602,281	618,037	647,021	1,625,068	0
Col 16	2,752,743	0	15*	488,613	392,395	363,490	402,925	1,105,305	0
Col 17	1,441,228	0	0	212,936	209,886	254,547	244,096	519,783	0
Col 18	0	0	0	0	0	0	0	0	0

Table 12: Targus Data Table with Suppressing Sensitive Cells